Investigation - Absorption and Radiation of Heat

Write up the following investigation using the standard Keynote or Pages.

INQUIRY: INVESTIGATION 10.3

Radiating and absorbing radiant heat

KEY INQUIRY SKILLS:

- questioning and predicting
- processing and analysing data and information

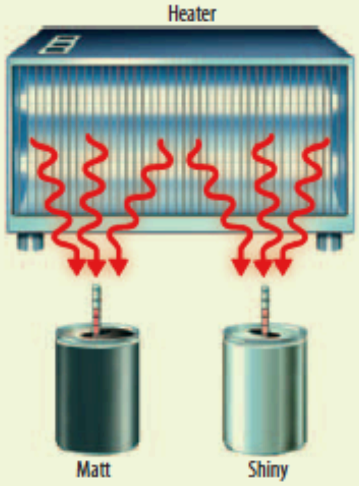
Equipment:

heater or microscope lamp

2 identical shiny, empty, soft-drink cans

matt black paint and paintbrush

2 thermometers or data logger and 2 temperature sensors



- Paint one of the cans matt black and leave the other as it is.
- Construct a table or spreadsheet headed 'Radiating heat' in which to record the temperature every two minutes for up to 14 minutes.
- Make a prediction about your results and write down a hypothesis.
- Pour equal amounts of hot water into each can.
- Place a thermometer or temperature probe into each can. Measure the initial temperature of the hot water and again every two minutes.
- Enter your data into the table or spreadsheet.
- Empty the cans and pour equal amounts of cold tap water into each can.
- Construct a table or spreadsheet headed 'Absorbing radiant heat' in which to record the temperature every two minutes for up to 14 minutes.
- Make a prediction about your results and write down a hypothesis.
- Place a thermometer or temperature probe into each can. Measure the initial temperature of the water.
- Place the two cans at equal distances from a heater or microscope lamp.
- Measure and record your data into the table or spreadsheet every two minutes.
- Plot line graphs that show how the temperature changed over 14 minutes during the cooling and heating of the cans. You may wish to plot both graphs on the same set of axes.

DISCUSS AND EXPLAIN

1. Which can radiated heat more quickly?
2. Which can absorbed heat more quickly?
3. Were each of your hypotheses supported by the data?
4. Why was it important to use cans that were identical in size and shape?
5. What other variables had to be controlled during this experiment?